Adarsh Kumar Jha

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EDUCATION

Indraprastha Institute of Information Technology Delhi (IIIT-Delhi) *2022 -- 2026*
B.Tech, Computer Science and Artificial Intelligence *CGPA: 8.2 / 10*

WORK EXPERIENCE

Software Engineer -- AI / Systems (Production Engineering) *Jan 2026 -- Present*
TBO.com (Travel Boutique Online) | Intern: Jan -- Jun 2026 → Full-Time: Jun 2026 -- Present

- Engineered a production **conversational AI** system using **AWS Nova Sonic** with **RAG** and tool calling; diagnosed latency regressions through **AWS CloudWatch**, sustaining **p95 latency under 600ms**.
- Orchestrated classification of **650+ daily customer emails** and **100,000+ notes** using **AWS Lambda**; applied prompt caching to slash inference spend by **70%+** and manual review by **57%**.
- Constructed distributed **ETL pipelines** using **Apache Spark**, **AWS Glue**, and **Kafka CDC (Debezium)**; stored lakehouse datasets in **Apache Hudi** on S3 across **billions of records**.

ML Engineer Intern *May 2025 -- Oct 2025*
IIIT-Delhi -- Research to Commercialization (R2C) | Advisor: Dr. Anuj Grover

- Optimized a Transformer QA model via **LoRA** and quantization, achieving **97% accuracy** while cutting model size by **75%** and doubling inference throughput on document parsing tasks.
- Integrated **OCR** into a **FastAPI** inference service, scaling to **100+ concurrent requests** and improving entity extraction accuracy by **23%**.

Independent Research Project -- Ontology Path Extraction *Aug 2024 -- Dec 2024*
IIIT-Delhi x Oracle | Advisors: Prof. Raghava Mutharaju, Suman Roy (Oracle)

- Fine-tuned **R-BERT** in **PyTorch** with span-based **embeddings**, surpassing LLaMA-3 and GPT-3 baselines by **6 F1 points**; indexed **25,000+ biomedical entities** via FAISS.

PROJECTS

VICAS Lab Portal & Resume Knowledge Graph | *React, FastAPI, FAISS, Neo4j, Docker* [Code] *Aug 2025 -- Apr 2026*

- Shipped a production portal with **AWS EC2**, **JWT + Google OAuth**, 6 REST modules, and CI/CD via **GitHub Actions**; onboarded **160+ registered students**.
- Built a **Gemini RAG** pipeline transforming resumes into structured candidate profiles in **FAISS**; trained a **GCN** for skill link prediction with cross-encoder re-ranking.

ML Deployment Debugger & Observability Dashboard | *Python, FastAPI, Docker, Kubernetes, GCP* [Code] *Jan 2025 -- Apr 2025*

- Created a diagnostics engine intercepting ML inference services on **Kubernetes**, isolating latency spikes and OOM faults to cut mean time to diagnose by **40%**.
- Deployed on **GCP** with Cloud Storage artifact logging and a **FastAPI** dashboard for real-time inference telemetry.

Watch-Tower: Edge-AI Video Anomaly Detection | *PyTorch, TensorRT, FastAPI, Docker, MQTT* [Code] *Aug 2024 -- Dec 2024*

- Pioneered a real-time edge anomaly detector with **PyTorch** and **TensorRT**, streaming inference over CCTV feeds via **MQTT** with **sub-50ms latency**.
- Containerized microservices using **Docker** and **FastAPI**, achieving **91.4% ROC-AUC** on benchmark video datasets.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL, Bash, Go
AI/ML: PyTorch, LLMs, Transformers, RAG, AI Agents, Tool Calling, MCP, LoRA, PEFT, FAISS, GNNs, Vector Search
Cloud & Systems: AWS (EC2, S3, Lambda, Glue, DynamoDB, CloudWatch, Nova Sonic), GCP, Docker, Kubernetes, Linux, Nginx
Backend & Data: FastAPI, Express.js, React, Neo4j, REST APIs, Git, CI/CD, Apache Spark, Apache Kafka, Apache Hudi, Parquet, Debezium

ACHIEVEMENTS & LEADERSHIP

Ranked AIR 252 in GATE 2026 (CS), demonstrating core computer science problem-solving fundamentals *2026*
Joint Entrance Examination Mains 2022: All India Rank **4778 (99.5 percentile)** among 872,970 candidates *2022*
Competitive Programming: Codeforces (Specialist - Max Rating 1450+), LeetCode 300+ problems, GFG *2023 -- 2026*
Operations Head, Odyssey IIIT-D -- Led 30-member logistics team for cultural festival hosting 30,000+ attendees *2025*